Starting Up AI

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CES 24-09 **March 2024**

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Abstract

Using comprehensive administrative data on business applications over the period 2004-2023, we study emerging business ideas for developing AI technologies or producing goods or services that use, integrate, or rely on AI. The annual number of new AI business applications is stable between 2004 and 2012 but begins to rise after 2012, and increases faster from 2016 onward into the pandemic, with a large, discrete jump in 2023. The distribution of AI business applications is highly uneven across states and sectors. AI business applications have a higher likelihood of becoming employer startups and higher expected initial employment compared to other business applications. Moreover, controlling for application characteristics, employer businesses originating from AI business applications exhibit higher employment, revenue, payroll, average pay per employee, and labor share, but have similar labor productivity and lower survival rate, compared to those originating from other business applications. While these early patterns may change as the diffusion of AI progresses, the rapid rise in AI business applications, combined with their generally higher rate of transition to employers and better performance in some post-transition outcomes, suggests a small but growing contribution from these applications to business dynamism.

* Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau or the U.S. Department of State. Nikolas Zolas was affiliated with the U.S. Census Bureau during the main analysis period for this paper. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed the results for disclosure avoidance protection (Project 7500391: CBDRB-FY24-ESMD001-001, CBDRB-FY24-0147). We thank David Beede, Richard Beem, Cathy Buffington, Tim Dunne, Lucia Foster, Nathan Goldschlag, Cheryl Grim, John Haltiwanger, Zachary Kroff, Neil Thompson, and participants at the MIT CSAIL FutureTech research seminar for comments and suggestions. Author contact information: emin.m.dinlersoz@census.gov; cdogan@radford.edu; ZolasNJ@state.gov.

1 Introduction

Artificial Intelligence (AI) is a collection of technologies that can perform tasks that require human intelligence, such as visual perception, speech recognition, and decision-making. For several years now, industry analysts, academics, and the popular press have been hypothesizing about how artificial intelligence will transform the future of work and productivity (see, e.g., Baily et al. (2023); Acemoglu and Restrepo (2020); Agrawal et al. (2019)). But how fast is AI diffusing among U.S. firms? Evidence from recent U.S. Census Bureau surveys suggests that AI use rate remains relatively low among firms (3-5% range for much of the period 2017-2023), and adoption is concentrated in large firms.[1] Nevertheless, there are now signs that AI diffusion is accelerating, as the recent estimates from the Census Bureau's Business Trends and Outlook Survey (BTOS) indicate that AI use rate among firms is nearly 5.5% as of late 2023, up from 3.2% for the period 2016-2018 based on the Annual Business Survey (ABS).[2] Consistent with this acceleration, forward-looking questions in BTOS on expected future AI use by businesses indicate that the use rate may reach as high as 8.5% by mid-2024.[3]

Why is AI adoption apparently low? Aside from the difficulties of measuring AI use in surveys and the resulting potential underestimation of AI adoption rate among firms, there are many reasons.[4] AI is still a nascent technology with growing applications to a diverse set of business problems. These applications can only be developed over time with specific technological breakthroughs in AI and related technologies – a recent example is the development of generative AI and its applications.[5] Moreover, many firms lack scale, complementary technologies, and relevant infrastructure, such as high computing power and

[1] See, e.g., McElheran et al. (2023); Zolas et al. (2020); Acemoglu et al. (2022).

[2] See Acemoglu et al. (2022), Breaux and Dinlersoz (2023), and https://www.census.gov/hfp/btos/data. The AI use rate from BTOS is adjusted to remove "don't know" answers to make it comparable to the ABS adoption rate. The unadjusted BTOS use rate is about 5%.

[3] For bi-weekly estimates of current and expected future AI use rates, visit BTOS - Data (census.gov). This figure reflects the adjustment to exclude "don't know" answers to the future adoption question. The unadjusted rate is about 7%.

[4] Difficulties in measuring AI can lead to measurement error in surveys. Many recent surveys, including those by the U.S. Census Bureau, focus on measuring the use of AI specifically as part of a production process for goods or services, or AI as part of a product or service provided by the firm. Many firms likely use AI only incidentally or insignificantly but not report such use, as AI may not be a part of their core processes or products. Others may rely on AI embedded in systems or services supplied by third-parties, in which case they are unknowing users of AI.

[5] Generative AI is a form of artificial intelligence that is capable of generating text, images, or other types of data based on generative models, often in response to a prompt in a user interface.

vast amounts of data in digital form, to benefit from some AI applications.[6] Thus, many small businesses, which represent the bulk of the firm population, do not adopt AI, and most may not yet find it readily applicable to their businesses – a pattern that may be changing with the advent of generative AI and its applications to business problems. The limited applicability and cost of AI are indeed often cited as key barriers to wider adoption and use among firms.[7]

Another reason, relevant for our focus in this paper, may be the need for organizational change to accommodate AI. AI diffusion is fueled by two margins: the entry of new businesses using or providing AI, and the adoption or provision of AI by existing businesses. Existing businesses can benefit from AI only to the extent they can change the way work is organized to utilize AI effectively. For example, productivity improvements from AI may materialize only when workers are retrained and existing processes and systems are upgraded. However, it may be harder for some existing businesses to do so due to a variety of frictions. This need for organizational transformation is one reason why it is sometimes easier for early-stage businesses to take advantage of emerging technologies, including AI, as they can readily integrate these technologies with business processes early on or even design the business around a new technology.[8] The diffusion of AI may thus already be accelerating among the population of early-stage or nascent businesses from which new AI-using startups will emerge in the near future.

In this paper, we provide new evidence on the prevalence of AI-related business ideas among potential entrepreneurs who seek to start new businesses. Instead of surveys, we rely on comprehensive administrative data on business applications from the U.S. Census Bureau's Business Formation Statistics (BFS) to identify AI-related business applications (henceforth, *AI business applications*) over the period 2004-2023, and employer startups originating from these applications. The BFS micro data is based on applications for Employer Identification Numbers (EINs) filed with the IRS using Form SS-4, which also collects information on the nature of the planned business activity. Using text analysis on an applicant's reported main business line, business name, and trade name in conjunction, we are able to identify business ideas for developing or providing AI technologies and applications, as well as those that aim to use AI in providing goods or services. Our approach based on

[6] See Zolas et al. (2020) for discussion on digital hierarchies.

[7] see evidence from the 2019 ABS in Acemoglu et al. (2022).

[8] This is consistent with the fact that young firms are more likely adopters of new and advanced technologies (including AI), conditional on other characteristics – see, e.g., Zolas et al. (2020); Acemoglu et al. (2022).

administrative data is complementary to surveys that measure AI adoption and use, and particularly informative on early-stage business activity not typically captured in surveys.

We find that new AI business applications arrive at a largely stable rate annually for the period 2005-2012. The first inflection point occurs around 2012, when the annual number of applications starts to increase, albeit slowly. 2016 appears to be yet another turning point, after which applications exhibit faster rise that continues into the Covid-19 Pandemic and beyond. In particular, we observe a large, discrete jump in application activity in 2023, potentially attributable to the emergence of generative AI that became more widely available starting in late 2022. We provide several pieces of external evidence in support of this observed time-path and its turning points. We also find a high degree of concentration in AI business applications across sectors and states over the analysis period: nearly 56% of all applications are in Professional, Scientific, and Technical Services (NAICS 54), and about 20% originate from California – consistent with the patterns of high sectoral and geographic concentration observed in earlier surveys; particularly in the Annual Business Survey (ABS) and the more recent Business Trends and Outlook Survey (BTOS).[9]

The annual number of employer startups originating from AI business applications is also increasing in recent years, driven by the rising number of applications and the fact that AI business applications have a higher than average rate of transitioning to employer businesses. Nevertheless, it may take some time for this rising AI startup activity to make a large dent in the overall AI adoption rate in the firm population. This is in part because the current levels of annual AI business applications are still small relative to the number of existing firms, and it takes time for startups to form from applications. It is also in part due to the fact that the overall adoption rate depends on the survival rate of these startups, as well as the rates of adoption and de-adoption by existing firms. However, as we discuss below, our approach likely underestimates the underlying count of AI business applications, and the contribution to the AI adoption rate from new business initiations could thus be larger.

We also examine the early characteristics of AI business applications and compare them with those of other applications. Each business application is accompanied by a set of characteristics submitted by the applicant, such as the reason for application, intent to hire employees, legal form of organization, previous EIN application, plans to make wage payments, and expected initial employment. Many of these characteristics may provide, mostly indirectly, information on the viability of the business idea. We show that AI busi-

[9]See Zolas et al. (2020); Acemoglu et al. (2022); McElheran et al. (2023).

ness applications have initial characteristics associated with a higher potential to become an employer business and generate superior post-transition outcomes, at least in some dimensions. We find that AI applications indeed turn into employer startups at a higher rate and expect higher initial employment. Furthermore, conditional on application characteristics, employer businesses that form from AI business applications tend to be larger in terms of employment and revenue, and have higher payroll, average pay per employee, and labor share in their early life-cycles, compared to employer businesses forming from other applications.[10] However, they exhibit similar labor productivity and lower survival rate.

We emphasize that our findings do not necessarily indicate causal links; they mainly describe the key features of AI business applications and startups. Moreover, the patterns we document pertain to the very early stages of the diffusion of AI and they can change as the diffusion progresses. Nevertheless, the recent acceleration in AI business applications and their higher rates of transition to employer businesses, along with the better performance of AI employer businesses in a number of key outcomes, suggest an increasing contribution of AI business initiation activity to business dynamism and growth in the future – especially in light of the fact that business applications with high propensity to become employers are very early indicators of future employer business formation and aggregate employment growth (see Asturias et al. (2021, 2023)).

As a by-product of our analysis, a future goal is to develop an experimental public data series on AI business applications and formations that could be updated monthly to provide high-frequency information on the diffusion of AI among early-stage businesses. This series, along with other data, can inform policymakers on AI's diffusion and effects, and offer insights for the business community into the nature of the emerging AI business activity. Our approach for identifying AI business ideas is also applicable to other advanced or emerging technologies, such as robotics, cloud computing, and others. In future work, we plan to combine information on business initiations related to several emerging technologies into a high-frequency index that tracks the technological sophistication of early-stage business activity in the United States.

The paper proceeds as follows. We lay out our conceptual framework in Section 2. Section 3 describes the data. Section 4 presents our findings. A brief summary of the results and a discussion are provided in section 5. Section 6 concludes.

[10] In related research using early-stage startups, McElheran et al. (2023) find that conditional on survival AI-using startups tend to exhibit faster revenue growth than others, consistent with our findings that AI startups perform better in some key outcomes.

2 Conceptual Framework

To conceptualize our measurement of AI business applications and formations, we consider a simple framework. In any period (year) t, there are N_t employer businesses in an economy, N_t^a of which use or provide AI - collectively referred to as AI businesses. Define the share of AI businesses as

$$a_t = \frac{N_t^a}{N_t}. \tag{1}$$

In the beginning of period $t+1$, businesses that have not yet used or provided AI start doing so with a rate r_{t+1}, and a fraction d_{t+1} of businesses that have used or provided AI in period t cease to do so. In addition, E_{t+1}^a new AI businesses enter during period $t+1$, while X_{t+1}^a of the existing AI businesses exit at the end of the period. As a result, the stock of AI businesses evolves according to

$$N_{t+1}^a = N_t^a(1-d_{t+1}) + r_{t+1}(N_t - N_t^a) + E_{t+1}^a - X_{t+1}^a. \tag{2}$$

The share of AI businesses at the end of period $t+1$ is then

$$a_{t+1} = \frac{N_t^a(1-d_{t+1}) + r_{t+1}(N_t - N_t^a) + E_{t+1}^a - X_{t+1}^a}{N_{t+1}}. \tag{3}$$

Dividing the numerator and the denominator through by N_t yields

$$a_{t+1} = \frac{a_t(1-d_{t+1}) + r_{t+1}(1-a_t) + (\varepsilon_{t+1}^a - \chi_{t+1}^a)}{1+g_{t+1}}, \tag{4}$$

where $g_{t+1} = (N_{t+1}/N_t) - 1$ is the growth rate of the total number of businesses in the economy and

$$\varepsilon_{t+1}^a = \frac{\varepsilon_{t+1}^a}{N_t}, \;\; \chi_{t+1}^a = \frac{x_{t+1}^a}{N_t} \tag{5}$$

are the entry and exit rates of AI businesses in period $t+1$, relative to the stock of all businesses at the end of period t.

Based on (4), the adoption rate is fueled by two key margins: the share of existing businesses that adopt, $r_{t+1}(1-a_t)$, and the entry rate of new AI businesses, ε_{t+1}^a. The entry rate plays a key role if the adoption rate of existing businesses is low. For instance, if existing businesses face large frictions in adoption, entry by new AI businesses would contribute more to the growth in adoption. The share of AI businesses increases from one period to another

$(a_{t+1} > a_t)$ if

$$\varepsilon^a_{t+1} > a_t g_{t+1} + d_{t+1} a_t + \chi^a_{t+1} - r_{t+1}(1 - a_t). \tag{6}$$

In other words, the rate of entry needs to be sufficiently high to surpass the current adoption rate adjusted by the overall growth in firm population, and overcome the de-adoption and exit rate of existing AI firms – net of the rate of increase in existing firms' AI use or provision.

While various Census Bureau surveys generated estimates of a_t for recent years, we lack reliable estimates of d_{t+1}, r_{t+1}, ε^a_{t+1}, and χ^a_{t+1}.[11] The focus of our empirical analysis is on the entry rate

$$\varepsilon^a_{t+1} = E^a_{t+1}/N_t = (I^a_{t+1}\tau_{t+1})/N_t = \rho_{t+1}\tau_{t+1}, \tag{7}$$

where I^a_{t+1} is the stock of AI business applications (or ideas) that can potentially transition into employer businesses at the beginning of period $t+1$, and τ_{t+1} is the transition rate of each such idea to an actual startup in period $t+1$. The entry rate is higher, the higher the ratio of AI business applications to the number of existing businesses, ρ_{t+1}, or the higher the transition rate of ideas, τ_{t+1}.

Note that the stock of AI business ideas at risk of entry can be written as $I^a_{t+1} = (1-\tau_t)I^a_t + F^a_{t+1}$, where F^a_{t+1} is the flow of new AI business ideas in period $t+1$. The empirical analysis first provides some estimates of the annual flow of AI business ideas (or applications), F^a_t, their transition rate, τ_t, and the resulting number of startups. In addition, we explore the characteristics of AI business applications to understand how they differ from other applications. Finally, we study the performance of employer businesses that originate from AI business applications in comparison to those that originate from all other business applications.

3 Data

3.1 Business Formation Statistics

The core analysis relies on micro data underlying the U.S. Census Bureau's Business Formations Statistics (BFS). Next, we describe key aspects of BFS relevant for our analysis – see also Bayard et al. (2018) and Dinlersoz et al. (2023) for background on BFS micro

[11] See Zolas et al. (2020); Acemoglu et al. (2022); McElheran et al. (2023) for recent estimates of a_t. Some recent estimates of d_{t+1} will be available from BTOS, which asks businesses about both their current and expected future use of AI.

data.

3.1.1 Business Applications

The BFS provides high-frequency statistics on early-stage entrepreneurial activity based on applications for employer identification numbers (EINs).[12] Applications for new EINs are received by the Census Bureau on a weekly basis from the IRS. An EIN application, made using IRS Form SS-4, contains information on a variety of application characteristics. Applications are also assigned detailed industry codes (6-digit NAICS) and geocoded. Based on this information, the BFS identifies a core set of EIN applications that are primarily made with an intent to start a business, by filtering out applications made for other reasons (e.g., for trusts, estates, and other financial filings), applications from public entities, applications originating outside of the U.S., and applications related to certain agricultural, financial, or private household activity.[13]

Employer businesses are required to have an EIN to file payroll taxes. Similarly, all businesses (employer or nonemployer) must file for an EIN to form a partnership or a corporation. The BFS data does not cover some business types that can operate without an EIN – mainly sole proprietor nonemployers. However, many of these businesses also need and, in fact, use an EIN; for example, to establish an official business bank account, to establish a more formal business identity, or to conform to certain IRS regulations (Fairlie et al. (2023)). While there are many sole proprietor nonemployers without EINs, they account for a small fraction of aggregate economic activity.[14] Thus, the BFS micro data on applications provides nearly full coverage of all economically significant business initiations.

The information on the application form includes the name and address of the applicant and business, application week, business start date, reason for application (hiring, starting a new business, purchasing an ongoing business, banking, etc.), type of business entity, previous application for an EIN, principal industry and main line of business, planned date for initial wage payments, and the highest number of employees expected in the first 12 months following application (the latter available after 2017). Some of these characteristics provide information on the underlying business idea and are key predictors of the likelihood

[12] For public-use BFS data products, see the BFS website.

[13] The details of the micro data and the construction of the BFS are in Bayard et al. (2018), including the exact set of filters used to exclude applications.

[14] Futhermore, Davis et al. (2009) find that nonemployers with an EIN (including sole proprietors) are on average much larger in terms of revenue than those without an EIN.

of transitioning to an employer business. Bayard et al. (2018) and the BFS program use these characteristics in probit or linear probability models (LPM) to predict the likelihood of transition and generate projected business formations for recent time periods for which actual formations are not yet available. This prior work finds that applications that indicate a planned wage payment date, applications from a corporate entity, applications that are made with a goal to hire or start a new business (versus other reasons, such as banking or conformity to IRS regulations), and applications in certain industries (such as retail and certain professional services) tend to transition with a higher likelihood.

Our focus is on business applications related to AI. Just like the rest of the applications, these applications can be intended for an employer or a nonemployer business – e.g., AI code developers, researchers, or consultants. However, for job creation and contribution to overall economic activity, applications that are more likely to transition to employer businesses are critical.[15] Using application characteristics, the BFS program identifies a subset of all applications as "high-propensity business applications" - also referred to as "HBA" in BFS. These are applications that (i) are made by a corporate entity, (ii) indicate hiring employees, (iii) provide a (planned) first wages-paid date, or (iv) have a NAICS industry code in specific sectors and industries – accommodation and food services (NAICS 72) or in portions of construction (NAICS 237, 238), manufacturing (NAICS 312, 321, 322, 332), retail (NAICS 44, 452), professional, scientific, and technical services (NAICS 5411, 5413), educational services (NAICS 6111), and health care (NAICS 621, 623).[16] High-propensity applications tend to transition to an employer status with a much higher likelihood (23.9%) than non-high propensity (5.3%) over the sample period. We therefore explore the extent to which AI business applications are "high-propensity", and whether they disproportionately possess certain component features of "high-propensity" applications - e.g., indicating planned wage payments or coming from corporations.

While business applications are available at a monthly or weekly frequency, for the purposes of understanding the broad evolution of AI business initiation activity in recent decades annual data is sufficient. The annual frequency also avoids the complications of higher noise and seasonality in the monthly and weekly data. However, in the future we plan to use monthly data (with seasonal adjustment) for an experimental public-use data product that tracks AI business applications over time at a high frequency – see our conclusion below.

[15] Asturias et al. (2021, 2023) show that applications that are more likely to transition to employers are especially early indicators of aggregate economic activity.

[16] See Business Formation Statistics - Methodology (census.gov) and Bayard et al. (2018).

For our analysis, we mainly focus on the period 2004q3-2023m10 – the 2004 data is only available starting in the third quarter, and the 2023 data contains all months until November. As we discuss below, this period covers years both before and after the key technological developments and milestones relevant for the recent surge in AI business ideas.

3.1.2 Business formations

BFS combines the confidential micro data underlying the EIN applications with the Longitudinal Business Database (LBD) – the longitudinal version of the Census Bureau's Business Register. The LBD contains firm and establishment-level information on age, location, industry, number of employees, quarterly payroll, and EIN for employer businesses. BFS uses the EIN to match business applications to the LBD to identify the incidence and timing of transitions to new employer businesses.

In documenting employer business formations (i.e., transitions or startups) from AI business applications, we focus on entirely new employer businesses (i.e. "age zero" firms in the LBD) that form within 8 quarters of the application date – nearly 90% percent of transitions to employer businesses occur within eight quarters of the application.[17] We construct the transition rate of applications to employer businesses as the ratio of the number of transitions within 8 quarters to the number of business applications these transitions originate from. Because the latest vintage of the LBD covers years up to 2021, we do not observe 8-quarter window formations from applications made after 2019. Thus, for the period 2020-2023 we use projected (or predicted) formations the BFS provides based on a linear probability model (LPM) that leverages the rich set of application characteristics as predictors.[18]

We also follow AI business applications after transition to explore whether AI startups systematically differ in business outcomes from other startups. In tracking post-transition performance of startups, we use all new employer businesses originating from applications without the 8-quarter window restriction – thus including all employer business formations until the end of 2021. As we detail below, in this analysis we control for initial application characteristics, as well as the startup life-cycle state (summarized by size and age), as needed.

Overall, the micro data contains more than 60 million applications and nearly 8 million employer startups since 2004. We use this rich data to identify AI business applications and startups based on a methodology which we describe next.

[17] See Figure 2 in Dinlersoz et al. (2023).

[18] See Bayard et al. (2018) for a description of the LPM model that the BFS program uses to track the evolution of predicted formations for periods for which the actuals are not yet available.

3.2 Identifying AI business applications

To identify AI business ideas or applications, we rely on three key variables in the BFS micro data obtained directly from EIN applications: main business line (principal line of merchandise), business name, and trade name. These variables contain the text written-in by an applicant, and provide information on the planned main product or service, and the name of the business. We search these three text fields in conjunction to isolate cases that contain key words related to AI technologies and applications.[19] Importantly, we do not restrict our search to certain industries with more intense reliance on AI (e.g., high-tech industries or professional services), since we want to capture AI business initiations in any sector as they emerge, to the extent possible.

As a first step, we form a comprehensive dictionary of AI-related keywords based on the accumulated AI literature and practice, and input from experts (Emerging Technology Fellows from "xD" at the Census Bureau).[20] This large dictionary spans several AI technologies and their applications, broadly and narrowly defined, such as artificial intelligence, machine learning, deep learning, computer/machine vision, image processing, speech/voice recognition, neural networks, large language models, virtual agents, generative AI, etc. It also includes several keywords on business implementations of AI technologies, such as robotics process automation, self-driving cars, AI art, etc. We supplement this dictionary with a list of additional keywords related to AI technologies and their applications provided by ChatGPT. We also add variations of several AI-related keywords to allow for slight deviations or spelling mistakes, common abbreviations, and closely related phrases and synonyms.

In the second step, we use the dictionary to identify business applications related to AI by searching the three text fields in the data described above and finding matches to the entries in the dictionary. Finally, we perform extensive filtering to exclude those applications that are not aimed at AI businesses. In filtering, we use all the information available in main business line, business name, and trade name in relation to each other to remove "false positives". If the business name or trade name for an application contains an AI keyword, we cross-check the main business line information to ensure that the cases unrelated to AI businesses are removed. For example, a business application is excluded if it has an AI-related keyword in its business name or trade name, but indicates a main business line

[19] Colaiacovo et al. (2023) use a similar approach, albeit based on firm names only, to identify whether a business registration in a state is associated with a particular technology.

[20] In particular, we thank Tomo Lazovich and Mike Walton for their input. To learn more about xD at Census, visit xD - U.S. Census Bureau.

unrelated to AI technologies or applications – such as the fictitious examples of "Superior AI Services" and "Ai Chung's Garden" with main business lines"artificial insemination for cattle" and "Chinese restaurant", respectively.

We also remove additional types of applications made for purposes not related to core AI business activity, such as hobbies, clubs, trade associations/shows, conventions/conferences, investment vehicles, and holding companies. However, the time trends in AI business applications, and the distributions of the applications across states and sectors, are highly robust to various filtering criteria we experimented with. Overall, our filtering results in a set of business applications that are closely related to the provision or use of AI technologies.

Table 1 shows the number of AI business applications we identify, along with the filtered number of applications. We break down the applications identified by each field of search (main business line, business name, trade name) – some applications may be in the intersection of two or all fields. A large number of AI business applications are identified based on main business line and business name, and a smaller subset based on trade name – mainly because a business is not required to have a trade name, unlike a business name and principal line of merchandise, which are available for almost all applications in our data. In some of the analysis to follow, we consider separately AI business applications based on main business line and the complement set that consists of AI business applications identified solely based on business/trade name to document any differences between the two groups. Overall, we end up with 21,731 AI business applications after filtering, 11,545 (53%) of which are identified based on main business line (and possibly on business/trade name), and the remaining 10,186 (47%) based on business/trade name only. Table 1 also shows that, perhaps not surprisingly, AI business applications constitute a very small fraction (0.035%) of the 62+ million business applications in our database as of late 2023.

AI business applications we identify include both the providers and users of AI technologies. Applications identified based on main business line tend to contain more AI providers and include those that report a main business line aimed to develop AI technologies, provide AI code/software, build or integrate AI systems, or offer AI consulting services - some of these are likely nonemployers. The ones identified based on business/trade name only are primarily users of AI technologies in their planned products or processes. However, both groups are ultimately a mix of both providers and users, and potential employers and nonemployers. For the current analysis, we do not attempt to classify applications into users or providers. In future work, we plan to use further techniques and other information in the application to

make that classification possible.[21] Overall, we view AI business applications in its entirety as an "ecosystem" for AI-related business activity – including users, innovators, developers, coders, consultants, and suppliers of AI.

3.3 Selection

While we are able to identify a large number of AI business applications, our approach depends on potential entrepreneurs' propensity to reveal the nature of their planned business by using an AI-related keyword in their main business line, business name, or trade name. Applications that contain such information may aim for businesses that provide AI, or that directly and significantly use AI. AI business applicants that see a only a supporting or less significant role for AI in their potential business may not reveal that information. Thus, our approach may be better suited for identifying applications where the potential entrant aims to develop or provide an AI technology, or uses AI in a relatively significant part of the planned process, product, or service. It may also be the case that more savvy AI entrepreneurs use AI-related keywords in their business names or descriptions, while others do not, despite having an AI-related business.[22] Our classification method is therefore subject to potential selection issues associated with reporting AI-related terms.[23] Similar selection issues are also encountered in surveys aiming to measure AI use by firms: non-response that is not random, incidental or insignificant use of AI that gets reported as no use, and unknowing use of AI that is not reported.

Because we cannot necessarily identify all underlying AI-related business activity based on the revealed textual information alone, the number of applications we end up with likely form a lower bound on the underlying true count of AI business applications. While we do not capture the exact underlying *level* of AI business applications, our approach is likely more informative on the *growth* in AI business applications. Below we rely on several approaches

[21] The information provided in the text fields we utilize is not always detailed enough to make that classification with precision. There are additional, non-text application characteristics that can be useful for classification – we show below that some of these characteristics are strongly associated with AI business applications. Future work will explore machine learning approaches to potentially improve the identification of AI business applications and their types – see, e.g., Miric et al. (2023) for a recent application of machine learning methods to the identification of AI patents.

[22] Some entrepreneurs may also be riding the "AI wave" by using catchy AI-related phrases to name businesses that use AI only incidentally.

[23] This selection is a more general challenge for the identification of any specific type of business activity in the BFS micro data, and not unique to the identification of AI business applications. See also Colaiacovo et al. (2023) for a discussion of similar issues in identifying businesses related to a given technology in state business registrations.

to externally validate the growth trends we observe.

We do not know the exact nature of the selection into reporting AI-related keywords with respect to the unobserved true set of AI business applications. Nevertheless, we can compare the observable characteristics of AI business applications we identify with those of all other business applications. We find that they tend to have higher transition likelihood to employer businesses, and exhibit better post-transition outcomes in certain dimensions. However, some of these findings appear to be mainly driven by observable characteristics of AI business applications, not necessarily by our methodology of classifying them as AI business applications per se. For some outcomes, once we control for observable application characteristics there remains little systematic relationship between the outcome and being classified as an AI application based on our methodology.

4 Findings

We first describe the evolution of AI business applications and actual employer businesses that form from these applications annually over the period 2005-2023. We then summarize the patterns of applications across states and sectors.

4.1 Time-path of AI business applications and formations

Figure 1 shows the annual number of AI business applications over time. Applications come in at a steady annual rate over the period 2005-2011, but the annual number of applications starts to increase in 2012 - the first turning point in the series. The second turning point occurs in 2016, after which there is a further acceleration in application activity that continues until 2022, including the Covid-19 pandemic period. A substantial, discrete jump in applications occurs in 2023. A similar time-path is observed for all AI business applications, those based on main business line, and high-propensity AI business applications.

Figure 1 also contains the annual number of employer startups that results from AI business applications within 8-quarters from the time of application. The time series contains actual startups for 2005-2019, and the projected startups for 2021-2023, for which actuals are not yet available. The projections are based on a linear probability model used by BFS that relates the event of startup formation to application characteristics.[24] By 2019, there

[24]See Bayard et al. (2018); Dinlersoz et al. (2021) for the details on the models used to estimate the likelihood of transition to an employer business within 8 quarters of the application.

were a total of about 1,300 startups that formed from AI applications (within 8 quarters of application) over the period since 2005, with much more (nearly 1,700) projected for the relatively short period 2020-2023 – consistent with the surge in applications in recent years. In general, all series follow similar time-paths and turning points, as shown in Figure A.1 in Appendix A, where each series is indexed to 100 for 2005. The patterns in Figures 1 and A.1 also generally conform with early phases of industry life-cycles during which entry escalates, and the early diffusion of innovations during which adoption rises – as documented in Gort and Klepper (1982), Rogers (1962), and the related literature.[25] We summarize our main finding in this section below.

Finding 1 *AI business applications and formations have been increasing rapidly in recent years.*

4.2 Characteristics of AI business applications

A key question is whether AI business applications systematically differ from other applications. In particular, do AI business applications *disproportionately* possess characteristics associated with a higher potential to create jobs? Do they differ from other applications in terms of their expected startup scale?

Table 2 indicates that around 42.9% of all non-AI business applications are high-propensity applications, which have on average a 23.9% share of transitioning into employer businesses as opposed to about 5.3% for non-high propensity applications. For AI business applications, the fraction that are high-propensity is 50.7% – about 18% higher (or 7.8pp) compared to non-AI applications. This fraction differs only slightly across the two AI application types: 51.3% for those identified based on main business line, and 50.8% for those identified based on business/trade name.

There is less of a separation in terms of indicating a first planned wage payment date (wage). 18.5% of non-AI business applications indicate planned wages, while 19.1% of AI business applications fall into this category – with modestly higher rates for applications based on main business line (20.3%) compared to 17.8% for business/trade name only. We note that applications with a planned wage date overall have a relatively high (37.4%) likelihood of transitioning into employer status, versus 7.8% for applications without planned wages.

[25] See, e.g., Jovanovic and MacDonald (1994), Klepper and Simons (2000) and Dinlersoz and MacDonald (2009).

Notably, AI business applications are disproportionately composed of applications made by corporate entities (corp). 19.2% of non-AI business applications are from corporations, whereas nearly 40% of AI business applications are from corporations, with similar rates for those identified based on main business line versus business/trade name only: 39.4% versus 40.5%, respectively. Applications from corporations have in general about 25.8% likelihood of transitioning into employer startups versus 10.3% for other legal forms of organization.

Next, we further unpack the connection between AI business applications and application characteristics using regression analysis. Table 3 contains the estimates from Linear Probability Models (LPMs). Unconditionally, a high-propensity application is associated with a 0.011 percentage points higher likelihood of being identified as an AI application (column 1) – equivalent to the 32% of the overall rate of AI application incidence (0.035%) among business applications. Conditional on detailed industry (4-digit NAICS), state, year, and application week (of the year) controls (column 2), a high-propensity application has 0.037pp higher likelihood of being an AI application (equivalent to 105% of the overall rate of AI application incidence).

Controlling further for more detailed application characteristics instead of the high-propensity indicator (column 3), we observe that many have a significant positive association with an AI application. These application characteristics are indicating planned wage payments (wage), being a corporate entity (corp) – versus the omitted category of other legal organization forms, indicating starting a new business or hiring an employee as a reason for application (new bus./hire) – versus the omitted category of other reasons for applying (such as banking or compliance with IRS), and having an executive or trade name (exec./trade name). In terms of the magnitudes, applications made by corporate entities stand out. An application made by a corporation (as opposed to other legal forms of organization) has nearly 0.05pp higher likelihood of being an AI application – equivalent to 140% of the overall rate of AI application incidence.

Doing the same analysis for AI business applications based on main business line versus business/trade name only results in qualitatively similar findings, but different magnitudes (columns 4-9). Our main finding is summarized below.

Finding 2 *AI business applications disproportionately possess characteristics associated with higher likelihood of becoming employer businesses.*

As an intensive measure of business application's prospects of job creation, we next consider the expected initial size of the planned business, measured by the highest (maximum)

number of employees expected in the first 12 months following the application, as reported by the applicants on the application form. This variable is only available from 2017 onwards in our sample. It takes on a value of zero if the applicant does not expect to hire in the first 12 months – e.g., if the applicant is aiming for a nonemployer business or does not plan to hire in the initial stages of an intended employer business. Otherwise, it is a positive integer that provides the applicant's subjective estimate of the maximum initial size of the intended business.[26] As shown in Table 2, the average expected size for non-AI business applications is about 0.62 employees – a figure dominated by the large presence of likely non-employer business applications. AI business applications, however, expect about 1 employee on average in their first year - nearly 60% higher than the average for non-AI applications. The expectations are even higher (1.15 employees) for AI business applications based on main business line, but lower (0.82 employees) for those based on business/trade name only.

Table 4 considers the relationship between application characteristics and the expected initial size using OLS regression. To accommodate the zeros, the dependent variable is the "TVV" or "DHS" transformation (instead of a log transformation) of the expected maximum number of employees.[27] Controlling for detailed industry, state, application year, and application week (column 1), an AI application is associated with 24% higher expected initial size.[28] The expectations are 29% higher for AI applications based on main business line, and 19% higher for AI applications based on business/trade name only (column 2). Once we control for application characteristics (columns 3 and 4), AI business applications have only 3.5% higher expected initial scale (about 4% higher for main business line cases and 3% for business/trade name cases). Including other application characteristics substantially attenuates the estimated coefficients for AI application indicators. In other words, the characteristics of the AI business applications that are associated with higher transition rates documented earlier accounts for most of the expected initial size premium associated with an AI application.[29] We summarize our main finding below.

[26]There is a small number of outliers in this variable that indicate initial expected maximum employment values over 500. To ensure that these outliers do not influence our analysis unduly, we winsorize this variable to 100 employees. The results are very similar when a lower threshold of 50 employees is used.

[27]Specifically, the transformation is $\widetilde{y} = 2\frac{(y-\bar{y})}{(y+\bar{y})}$, where $\bar{y}$ is the grand mean of the expected initial size in the entire sample—for further information on this transformation, see Tornqvist et al. (1985) and Davis et al. (1996).

[28]The coefficient estimates are converted to percentages by reverse-transforming the DHS transformation into a percentage difference. For each coefficient, this percentage difference is $100 \times 2\beta/(2-\beta)$.

[29]Note, in particular, applications with planned wages (wage) tend to be associated with a much larger expected initial size. Among applications, indicating planned wages is highly correlated with stating a positive expected initial size, and including this indicator (and other application characteristics) explains a

Finding 3 *AI business applicants anticipate higher initial scale (employment) than non-AI business applicants.*

The analysis so far indicates that AI business applications disproportionately have characteristics that are positively associated with a higher potential for job creation. These findings are consistent with survey-based evidence in McElheran et al. (2023) that AI-using young firms (0-5 year old employer startups) have early characteristics associated with high-growth entrepreneurship. Next, we examine the actual transition rates of AI business applications to employer businesses.

4.3 Transition rates and average duration to transition

As shown in Table 2, the transition rate (with 8 quarters of application) of non-AI business applications to employer startups over the period 2004q3-2019 is 13.3%. AI business applications have 1.5pp (or 11.2%) higher transition rate at 14.8%. The transition rates for AI business applications based on main business line versus business/trade name are 14.0% and 15.8%, respectively. We also observe that during the period 2004q3-2019, the transition rate of AI business applications is stable, whereas the transition rate of other applications drift persistently lower to about 8%, after a large drop at the beginning of the Great Recession – thus, the AI/non-AI business application transition rate gap is increasing. Overall, the higher transition rate of AI business applications is consistent with generally better transition potential of these applications based on their characteristics, as explored in the previous section.

Table 5 contains the estimates from LPMs that examine the association between the transition rate of AI business applications and application characteristics. AI business applications have about 2.9pp higher transition rate, controlling for state, detailed industry, state, application year, and application week (column 1). This amounts to a 22% higher rate compared to the transition rate of all applications (13.3%). The transition rate is higher by 2.5pp (or 19% relative to the mean) for AI business applications based on main business line, versus by 3.4pp (or 26% relative to the mean) for those based on business/trade name only (column 2). However, once other observable application characteristics are controlled for, AI business applications appear to be no different statistically in terms of transition rate from the rest of the applications (column 3). In fact, AI business applications based on main

substantial fraction of the variation in expected initial size.

business line have a slightly lower (by 1.2pp, or 9.0% relative to the mean) transition rate (column 4). In other words, the higher transition rate of AI business applications appear to be accounted for mainly by their observable characteristics that are correlated with a higher likelihood of transitioning – not by our methodology of designating them as AI business applications.

Finding 4 *AI business applications are more likely to transition into employer businesses than non-AI business applications, mainly explained by their initial characteristics.*

There are several frictions associated with starting a business. Some of these frictions may be reflected in the duration of the pre-transition gestational state that a potential entrant sojourns in after making a business application. The intended business may materialize shortly after the application or may take longer depending on the severity of the frictions as well as the nature and complexity of the business idea. Table 2 indicates that the average duration from application to transition, conditional on becoming an employer business within 8 quarters of application, is 1.77 quarters for non-AI applications and 2.15 quarters for AI applications. The average duration is slightly higher (2.18 quarters) for AI business applications based on main business line, and slightly lower (2.13 quarters) for those based on business/trade name only. In other words, AI business applications take slightly longer on average (by about 0.4 quarter) to transition.

Table 6 relates the DHS-transformed duration measure to observable characteristics of an application. Once we control for industry, state, application year and week, and other application characteristics, AI business applications do not appear to have a statistically different duration. Note also that applications indicating a planned wage date generally take less time to transition, whereas applications from corporations and those aiming to hire or start a new business take longer to transition, all else constant. Overall, while AI business ideas take slightly longer to implement, they are not particularly tardy once their initial characteristics are controlled for.

Finding 5 *Conditional on becoming an employer business, AI business applications take longer to transition than non-AI business applications, mainly explained by their initial characteristics.*

4.4 The distribution of AI business applications across states

We now examine the distribution of AI business application activity across states. Figure 2 maps the percent of AI business applications originating from each state for the entire period 2004q3-2023m10. The states with the largest shares are California (19.8%), Florida (9.4%), New York (8.3%), and Texas (7.5%). The distributions across states are similar for AI business applications based on main line of business versus business/trade name – see Appendix Figure A.2. Note that, some smaller states, such as Nevada, Utah, Delaware, and Wyoming appear to have disproportionate share of applications compared to their size. To see the concentration of AI business applications relative to the state size, we next normalize AI business applications by state population.

Figure 3 shows the number of AI business applications per capita (per 100,000 residents of age 18-64) based on the average population of each state over the period 2005-2022.[30] The states with the highest numbers of applications per capita are, by far, Delaware and Wyoming. Delaware stands out because AI business applications are disproportionately composed of applications from corporations, and Delaware receives a disproportionately high number of corporation applications owing to the favorable state corporate tax laws. Wyoming was named a federal tech hub and has recently added new tech incubators (such as IMPACT 307), and it has attracted blockchain/AI-related entrepreneurship in recent years.[31] These states are followed by District of Columbia, Nevada, California, Florida, and Utah. The rest of the states have much smaller AI business applications per capita. The patterns are similar for AI business applications based on main line of business versus business/trade name – see Appendix Figure A.3.

Finding 6 *AI business application activity is highly concentrated across states, with California having the highest share of applications.*

4.5 The distribution of AI business applications across sectors

Turning to the sectoral composition of AI business applications, we observe a high level of concentration in certain sectors. The share of AI business applications by sector are shown in Figure 4. The sectoral distribution is far more concentrated than the distribution across states explored in the previous section. In fact, a single sector stands out. A large share (56%)

[30] State-level population estimates by age are from the Census Bureau.
[31] See, e.g., 31 regional tech hubs and U of Wyoming tech hub.

of the applications is in Professional, Scientific, and Technical Services (NAICS 54). This is followed by Finance and Insurance (NAICS 52) and Information (NAICS 51), which contain 9.4% and 9.1% of the applications, respectively. The high concentration in NAICS 54 is also part of the reason why AI business applications tend to be disproportionately high propensity applications, since NAICS 54 contains two industries (NAICS 5411 - Legal Services and NAICS 5413 - Architectural, Engineering, and Related Services) that are classified as part of high-propensity applications.

Other notable sectors are Administrative and Support (NAICS 56), Retail (NAICS 44-45), Health Care (NAICS 62), Manufacturing (NAICS 31-33) and Other Services (NAICS 81) – each of these contain around 2-3% of AI business applications. Overall, NAICS 51 through 56 account for almost 80% of AI business applications – 82.3% of applications based on main business line and 74.9% of applications based on business/trade name. These sectoral patterns may change, however, as diverse applications of AI technologies continue to emerge in other sectors (such as arts and entertainment, NAICS 71). This sectoral distribution is also consistent with the sectoral distribution of AI-using firms observed in the 2019 ABS – see Acemoglu et al. (2022).

Finding 7 *AI business application activity across sectors is highly concentrated, particularly in Professional, Scientific, and Technical Services (NAICS 54).*

4.6 External validation and supporting evidence

We now perform a number of exercises to provide support for the evolution of AI business application activity documented so far.

4.6.1 Timing with respect to the milestones in recent AI history

The patterns of growth in AI business activity and the turning points in the time series are consistent with the timing of key developments in AI technologies in the last two decades. The relatively tranquil period (2005-2011) in AI business application activity saw a few key early developments that set the stage for major breakthroughs later in the 2010s. For example, in 2005, Stanford University's autonomous vehicle, Stanley, demonstrated the potential of AI in autonomous driving. In 2006, IBM Watson completed initial tests for competing in Jeopardy, eventually winning in 2011. The early methods to scale unsupervised learning to

big data were also developed during this period.[32] There were also notable advances in 2011 in speech recognition and computer vision – e.g., the release of Siri by Apple and Kinect for Xbox 360 by Microsoft.[33]

2012 was a turning point in AI history. Krizhevsky et al. (2012) introduced a deep convolutional neural network that won the ImageNet challenge for image classification and led to the explosion of deep learning research. Over the next few years, many other approaches to image recognition were abandoned in favor of deep learning, and other deep neural networks were developed.[34] These networks offered outstanding performance for traditional machine classification algorithms, enabling further applications of deep learning. Perhaps not coincidentally, 2012 is also the first turning point in the AI business applications series: the annual flow of new business applications starts to increase, albeit slowly.

Developments in AI continued in 2013, as Word2vec by Google was introduced to automatically identify semantic relationships between words, a key development in natural language processing.[35] In 2014, Goodfellow et al. (2014) invented generative adversarial networks, a class of machine learning approaches used to generate photos and transform images. Kingma et al. (2014) used variational methods to make generative approaches to supervised learning more efficient in large scale applications. Facebook (now Meta) developed the deep learning facial recognition system, DeepFace, to identify human faces in digital images.

2016 saw another major breakthrough with key developments in unsupervised and reinforcement learning. Notably, AlphaGo was able to play the game of Go at the level of top players, and make moves that these players could not even consider. Further advances in the ability of AI to imitate human behavior also took place in 2016. AI algorithms were developed to create text that mimics handwriting, and to imitate human speech (e.g., Google's Wavenet). Another significant development was Uber's initiation of a self-driving car pilot program in Pittsburgh. These advances in 2016 synchronize with the second turning point in AI business application activity. Business applications started to arrive at a much higher rate compared to the period 2012-2015. The applications of deep, unsupervised, and reinforcement learning methods to business problems undoubtedly propelled some of this growth.[36]

[32]For instance, Raina et al. (2009) presented the idea of using Graphics Processing Units (GPUs) to train large neural networks.

[33]The latter featured a machine learning application for human motion capture via computer vision.

[34]These include the Visual Geometry Group Very Deep Convolutional Network (VGGNet) and GoogLeNet.

[35]See, e.g., Mikolov et al. (2013).

[36]These applications include, but are not limited to, customer service and experience, quality control, man-

The capabilities of AI continued to improve after 2016. In 2017, researchers at Google introduced the concept of "transformers", inspiring the development of tools that could automatically parse unlabeled text into large language models (LLMs).[37] Advances enabled by transformer networks led to the release of the first generative pre-trained transformer (GPT-1) by OpenAI in 2018, followed in 2019 by GPT-2, which demonstrated the ability to generalize unsupervised to many different tasks. Meanwhile, on the medical front Google AI and Langone Medical Center's deep learning algorithm outperformed radiologists in detecting potential lung cancers.

The period 2020-2023 witnessed major advances in generative AI. In 2020, OpenAI released GPT-3, which is capable of generating human-like text. In 2021, OpenAI introduced Dall-E, an AI system that can generate images from text prompts. DallE contributed to the emergence of high-quality "AI art" from natural language prompts. Open AI's ChatGPT was released in November 2022 to provide a chat-based interface to its GPT-3.5. In 2023, OpenAI announced GPT-4, which receives both text and image prompts. Google also released its own chatbot (Bard), and Microsoft launched Bing Chat (later renamed Microsoft Copilot), a GPT-4-based chatbot integrated into the search engine.

The advances in generative AI in the early 2020s have broadened the applications of AI in business.[38] Perhaps not coincidentally, there is a discrete and substantial jump in AI business applications in 2023. The acceleration in AI business initiation activity during 2020-2023 was also in part driven by the Covid pandemic that increased incentives for automation more broadly. The rapid rise in AI business applications in recent years is also consistent with rising AI-related publications, conferences, courses, and other educational programs.[39] Overall, the turning points and the patterns of growth in AI business application time-series align well with the timing of the development of AI technologies with broader applications to business.

ufacturing safety, predictive maintenance, fraud detection, stock trading and investment, pricing strategies, marketing automation, and medical diagnosis.

[37] See Vaswani et al. (2017).

[38] Some of these applications are more sophisticated customer service, marketing automation, translation services and multilingual customer support, content creation, sentiment analysis and audience research, survey design, document review and analysis, compliance assistance, automated reporting, competitor intelligence, and the design of product catalogs and descriptions.

[39] For an extensive documentation of these trends, see Artifical Intelligence Index Report 2023.

4.6.2 Timing with respect to developments in complementary technologies

The time-path of AI business activity also aligns with the timing of the development of complementary technologies enabling AI. Advances in digitization, big data, cloud storage and computing, and computing power during the last two decades boosted AI development. These advances include the introduction of Simple Storage Service (S3) and Elastic Compute Cloud (EC2) in 2006 by Amazon's AWS, followed by the launch of Google App Engine in 2008, Microsoft Azure in 2010, and Microsoft Office SaaS on 2011. AI capabilities have been increasingly integrated with cloud services, enabling new possibilities for automation, analytics, and decision making. Similarly, increases in computing power and speed have enabled the implementation of large scale AI systems.[40]

Consistent with the data requirements and computing demands of AI, recent analysis using Census Bureau surveys (McElheran et al. (2023) and Zolas et al. (2020)) also shows that AI-using firms tend to have higher reliance on digitization and cloud services. The advances in these complementary technologies fueled the boom in AI business initiations.

4.6.3 Timing with respect to AI-related patents and trademarks

Additional validation of the time series for AI business applications comes from the data on intellectual property (IP) related to AI. Innovations in AI technologies by both existing and new businesses as reflected in AI-related patents can be a precursor of AI business applications. Furthermore, many of the emerging AI businesses need to protect their innovations, products, services, processes, and brand names by obtaining patents and trademarks early in their life-cycles. An escalation in the IP activity is thus expected to accompany the rise in AI business applications.

For AI-related patents, we use the AI patents identified by Branstetter et al. (2023) with a high confidence score (95%). AI-related trademark applications and registrations were identified by searching the trademark textual information in USPTO's Trademark Case File Database (TCFD) for AI-related keywords based on the same dictionary that was used to identify AI business applications – hence similar caveats apply as in our identification of AI applications.[41] Note also that AI patents and trademarks include IP activity by existing

[40] More powerful microchips to meet the processing demands of AI have been introduced recently, with the primary example being Nvidia's A100 GPU released in 2020.

[41] That is, a business that applies for a trademark for the name of a business or product related to AI may not necessarily mention an AI-related keyword in the trademark filing. While the dictionary used is the same, the underlying text in trademark applications is not the same source as the text in BFS micro data.

firms, not just by new businesses.

The evolution of AI business initiation is overall consistent with the time-path of AI-related patent filings and trademark activity (trademark applications and registrations), as shown in Figure 5. Over the period 2005-2018, the contemporaneous correlation between annual AI applications and AI patents granted is 0.78. The correlation between annual AI applications and AI trademark applications (registrations) is 0.98 (0.97). Figure A.4 shows a normalized version, where each series is indexed to 100 for 2005 for easier comparison of trends. AI business application activity appears to lag AI-related patenting considerably. AI business applications seem to have a shorter lag with AI-related trademark applications and trademark registrations. Trademark applications and registrations start to take off around 2010, about two years before AI business applications start to rise. The contemporaneous correlation of the annual growth rate of AI business applications with that of AI trademark applications (registrations) is 0.49 (0.52) over the period 2006-2018. In general, trademark applications are an early indicator (see deGrazia et al. (2020)), and registrations may have an even shorter lag with business formation activity since a registration requires proof of use of the trademark in commerce. Overall, the patterns in AI-related patenting and trademarking activity generally lend further support to the time-path of the AI business application activity, which appears to lag the IP activity.

4.6.4 AI adoption patterns in recent Census Bureau surveys

Recent evidence from Census Bureau surveys measuring AI use and adoption indicates a rise in the adoption rate. The use rate for the period 2016-2018 is estimated at 3.2% based on the 2019 ABS.[42] Most recent high-frequency measures of AI use rates in BTOS, on the other hand, indicate estimates close to 5.5% as of late 2023 – a 72% increase over the last 5-years.[43]

The increasing AI business applications and formations we document is consistent with the growth in adoption rate in recent years. Business applications are an early indicator of new business formation and non-farm employment growth.[44] The continuing rise in AI

Thus, the identification in the trademark data can be viewed as independent of the identification in the BFS micro data, conditional on a common dictionary.

[42]See Acemoglu et al. (2022). Zolas et al. (2020) also provides estimates of adoption rates in 2017 for some AI-related technologies based on the 2018 ABS.

[43]Estimates we reported based on BTOS public data exclude "don't know" answers to current and future AI adoption questions to be consistent with the estimates generated earlier based on the 2019 ABS, which exclude "don't know" answers.

[44]See Asturias et al. (2021, 2023).

business applications therefore suggests an increasing contribution to the adoption rate of AI by new AI startups in the near to medium term. This is particularly so because our methodology likely underestimates the underlying level of AI business applications and formations. Consistently, the future (6-month ahead) use rate of AI, measured in the BTOS as of the end of 2023, is expected to be as high as 8.5% by mid-2024. Furthermore, based on BTOS, AI use among the group of smallest firms (1-4 employees), which includes many startups, also appears to be accelerating: 6.4% as of late 2023, and expected to increase to 9.0% by mid 2024.

The sectoral and geographic distribution of AI business application activity also largely conforms to the patterns in recent Census Bureau surveys. AI use is most common in NAICS sectors 51, 52, and 54 based on the analysis of the 2019 ABS.[45] More recent estimates of the sectoral AI use rates from BTOS indicate that NAICS sectors 51-56 exhibit high rates; sector 51 leads, followed by sector 54. AI business application activity we document is also concentrated in these sectors, albeit with a much heavier presence in sector 54 compared to ABS and BTOS, since most applications developing or providing AI technologies likely fall into these sectors. Similarly, many states with high per capita AI business applications, such as California, Florida, Nevada, Utah, are also among the ones with relatively high rates of AI use among firms based on BTOS.

4.7 Performance of AI businesses

We now turn to the outcomes of employer businesses that originate from AI business applications at any point in time during the period 2004q3-2021. We compare the performance of these businesses over their life-cycles with that of the businesses originating from non-AI business applications.[46] Many AI businesses form after 2016 when AI business applications start to arrive at a higher rate. Since our outcome measures are available until 2021 (the last year for which the LBD is available), we are effectively documenting the early life-cycles (up to 5 years after birth) for many of the AI businesses – we have a longer horizon for AI

[45] See Acemoglu et al. (2022).

[46] For post-transition outcomes, we follow an application after it is first observed as an employer business in the LBD until the point the firm identifier associated with the business ceases to exist or a transition to multi-unit firm occurs. Almost all employer businesses in our sample are single-unit firms at birth, and we follow them as single-unit (single-establishment) firms over time. To be conservative, we stop tracking a startup if it becomes part of a multi-unit firm, because such transitions can be associated with M&A activity, in addition to natural expansion to a multi-establishment firm. We plan to distinguish more precisely between these two outcomes in future work.

businesses that formed earlier in the sample period.

Table 7 contains the OLS regression estimates that relate several key firm outcomes to application characteristics in a panel that tracks startups over time after formation. All dependent variables are DHS-transformed, except for the indicator of exit.[47] First, consider the firm size measures in columns 1-3. Conditional on a large number of fixed effects and initial application characteristics, AI businesses tend to exhibit 14% higher employment, 37% higher payroll, and 10% higher revenue.[48] Note also that initial application characteristics have significant positive relationships with firm size measures. For instance, indicating a planned wage payment date is associated with a 17% higher employment, and applications from corporations are associated with nearly 5% higher employment. Similarly, when the reason for application is to hire or start a new business, employment is higher by about 4%. Having a previous EIN and an executive or trade name are associated with 13% and 7% higher employment, respectively. Overall, initial characteristics tend to be significantly related to the scale of businesses originating from the applications.[49]

Because we mainly document the early-life cycles for most of the AI businesses, they may not exhibit exceptionally high profitability and productivity during these early years. This would be the case if they are primarily engaged, in these early years, in developing new products and processes integrating AI technologies without necessarily generating high revenue initially relative to their input costs. Table 7 relates the annual measures of average payroll (payroll/employment), labor share (payroll/revenue) and labor productivity (revenue/employment) outcomes to initial application characteristics (columns 4-6). Conditional on fixed effects, age and size classes, and initial application characteristics, AI businesses have 25% higher average payroll and 23% higher labor share compared to others. The larger average pay and labor share we observe is likely attributable to the generally higher skill level of AI businesses' workforce. Consistently, Acemoglu et al. (2022) find that firms indicate an increase in the overall skill of their employees due to AI use. However, we do not observe a significantly higher labor productivity for AI businesses at this early stage of the AI diffusion – consistent with their higher employment but not as much higher revenue (columns 1 and 3).

[47] All dependent variables in Table 7, except for the exit indicator, were winsorized at 99.9% to reduce the influence of some gross outliers.

[48] These estimates are obtained by converting the estimated coefficients from DHS transformation to percent differences.

[49] See also Table 9 in Bayard et al. (2018) for descriptive relationships between application characteristics and initial employment of startups.

Finally, the last column of Table 7 examines the relationship between business exit (failure) in any given year and application characteristics.[50] Conditional on controls, AI businesses exhibit about 3pp higher exit rate, or about 25% higher compared to the average annual exit rate of nearly 12% in the overall sample. We summarize below our key findings in this section.

Finding 8 *Controlling for application characteristics, AI businesses have higher employment, revenue, labor share, and average pay, but not higher labor productivity compared to non-AI businesses. They also exhibit higher exit rates.*

5 Summary and Discussion

This paper has documented some key facts about early-stage AI-related business activity. The findings point to a surge in AI business applications in the last decade, with particularly rapid growth in recent years. AI business applications are on average more likely to transition to employer businesses, and have better post-transition outcomes in some dimensions compared to other employer businesses – e.g. employment, pay, and revenue. AI businesses originating from applications appear to form a dynamic segment of employer business population in terms of turnover (entry and exit). This dynamic view is further supported by recent research that finds that AI-using startups tend to exhibit faster growth early on in their life-cycles compared to other startups – see McElheran et al. (2023).

While emerging AI businesses do not exhibit higher labor productivity compared to other businesses that form during the same period, this pattern may change. The higher labor share and average pay of these businesses suggest an early stage during which many AI startups may engage in developing processes or products by relying more heavily on high-skilled labor that leads to a higher labor cost relative to revenue, and prevents high labor productivity.[51] Labor productivity may rise over time relative to other businesses, as some surviving AI startups grow and generate increasing revenue relative to their labor input. In support of this possibility, Acemoglu et al. (2022) find, based on the 2019 ABS, that labor productivity increases with firm size, and AI-using firms as a whole (including larger

[50] Exit is defined using the Business Dynamics Statistics' (BDS) definition of exit based on employment. A single-unit firm exits if its employment is positive but becomes zero in the next period.

[51] Note that, by definition, a firm's labor productivity (revenue/employment) is the product of its labor share (payroll/revenue) and its average payroll (payroll/employment).

and older firms that use AI, not just young or small firms) have about 2% higher labor productivity conditional on firm characteristics.

Overall, the emerging landscape for AI business applications and startups suggests a small but growing contribution of AI business application activity to job-creating entrepreneurship and business dynamism. This optimistic picture, however, has to be interpreted with caution from the perspective of industry life-cycles and the diffusion of innovations. We are still at an early stage of the emergence of an "AI industry" in the U.S. – providers and users of AI all included. The broad literature on industry life-cycles and the diffusion of innovations suggests that the initial accelerated entry and adoption may well slow down or come to an end. A shakeout may follow at least for providers and users of some types of AI applications/technologies. For instance, the rapidly increasing demand for data and computation power for AI training can become, if not met with an adequately growing supply, a constraint that impedes further entry and trigger a shakeout – see, e.g., Bessen et al. (2022), Lohn and Musser (2022), and Lohn (2023). Technological constraints, data requirements, and emerging regulatory environment will all contribute to this evolution. At a broader level, the diffusion of AI will ultimately be a function of the distinct benefits and costs of AI for businesses, consumers, and workers, and how these benefits and costs change over time – see, e.g., Agrawal et al. (2023a,b); Svanberg et al. (2024).

6 Conclusion

Despite the growing interest in the rise of AI and its effects on businesses and workers, information on the nature of its diffusion among businesses has been limited. We provide new evidence on this front by using administrative data on business applications to identify emerging AI business ideas and startups originating from these ideas over the period 2004-2023. Our methodology relies on textual information revealed by applicants about their products and services, as well as the names of their businesses. While our approach does not necessarily pin down the *number* of all business applications aiming to develop or use AI, the *growth* patterns we observe are informative about the acceleration in new business activity related to AI, as several validation exercises suggest.

We document a sharp rise in AI business applications in recent years that is likely to persist into the near future as new AI technologies with applications to business processes continue to emerge. We find that AI business applications are associated with high potential

entrepreneurship. The higher transition rate of AI business applications and the generally better performance of resulting startups in certain business outcomes suggest a currently small, but growing contribution to business dynamism. The analysis also demonstrates the usefulness of BFS micro data in analyzing the emergence of new technologies and the evolution of new industries. An initial accelerated entry phase is an important stage of this evolution. BFS micro data can be used for early detection and analysis of this phase.

Our analysis also forms the genesis of a potential experimental public-use data product that can provide high-frequency information on the diffusion of AI among early-stage businesses. For example, a monthly series of new AI business applications can be released as part for the Census Bureau's BFS program. The keywords used to identify AI business applications can be updated over time to keep this series current and relevant as new AI technologies/applications appear. This experimental product can be a component of a broader ongoing effort that aims to measure various aspects of the diffusion of AI and its impact on the economy, providing additional insight for policymakers and businesses. Furthermore, the methodology for identifying AI business applications can be applied to other emerging technologies, opening up the possibility of building a high-frequency index of the diffusion of emerging technologies among nascent entrepreneurs.

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Table 1: AI Business Applications

Type of Identification	# of AI Applications	% of All AI Applications	% of All Applications
Unfiltered (Raw) AI applications			
AI - Main Business Line	11,970	43.7%	0.019%
AI - Business Name	15,872	57.9%	0.026%
AI - Trade Name	1,699	6.2%	0.003%
AI - All	27,370	100%	0.044%
Applications filtered out	5,639	20.6%	0.009%
Filtered (Core) AI applications			
AI - Main Business Line	11,545	53.1%	0.019%
AI - Business/Trade Name	10,186	46.9%	0.016%
AI - All	21,731	100%	0.035%

Notes: Includes all business applications for the period 2004q3-2023m10. For unfiltered (raw) AI business applications, percentages of all AI business applications do not add up to 100% because some applications can be in two or more identification types (e.g., main business line and trade name). For filtered (core) AI business applications, applications based on business or trade name (AI - Business/Trade Name) are the complement set of applications based on main business line (AI - Main Business Line).

Table 2: Characteristics of AI Business Applications

	Non-AI	AI		
		All	Main Bus. Line	Bus./Trade Name
% high propensity	42.9%	50.7%	51.3%	50.8%
% wage	18.5%	19.1%	20.3%	17.8%
% corp	19.2%	39.9%	39.4%	40.5%
avg. exp. max initial emp.	0.62	1.00	1.15	0.82
transition rate (8 qtr. window)	13.3%	14.8%	14.0%	15.8%
avg. duration (qtrs)	1.77	2.15	2.18	2.13

Notes: The analysis covers business applications for the period 2004q3-2023m10 and employer transitions for the period 2004q3-2019m12. "AI All", "AI Main Bus. Line", "AI Bus./Trade Name" indicate, respectively, an AI business application, an AI application based on main business line, and an AI application based on business/trade name. Non-AI indicates applications not classified as an AI business application. The application types are high-propensity business applications (high propensity), applications with planned wages (wage), and applications from corporations (corp). "Avg. exp. max initial emp." is the average expected maximum employment in the 12 months following the application. Transition rate is the fraction of applications that transition to an employer startup within a window of 8 quarters from the application. Avg. duration is the average duration measured in quarters (qtrs) from the quarter of application to the quarter of transition, conditional on transitioning within 8 quarters.

Table 3: AI business applications and application characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	AI	AI	AI	AI	AI	AI	AI	AI	AI
	All	All	All	Main Bus. Line	Main Bus. Line	Main Bus. Line	Bus./Trade Name	Bus./Trade Name	Bus./Trade Name
high propensity	0.0111***	0.0369***		0.0063***	0.0195***		0.0049***	0.0174***	
	(0.0005)	(0.0008)		(0.0004)	(0.0006)		(0.0003)	(0.0005)	
wage			0.0096***			0.0071***			0.0025***
			(0.0007)			(0.0005)			(0.0004)
corp			0.0486***			0.0246***			0.0240***
			(0.0009)			(0.0007)			(0.0006)
new bus./hire			0.0105***			0.0052***			0.0053***
			(0.0006)			(0.0005)			(0.0004)
prev. ein			-0.0012			-0.0016			0.0004
			(0.0014)			(0.0010)			(0.0009)
exec./trade name			0.0070***			0.0031***			0.0039***
			(0.0007)			(0.0005)			(0.0005)
bus. address			-0.0006			-0.0008			0.0002
			(0.0007)			(0.0005)			(0.0005)
year controls		*	*		*	*		*	*
state controls		*	*		*	*		*	*
week controls		*	*		*	*		*	*
industry controls		*	*		*	*		*	*
R^2	0.000009	0.00349	0.00352	0.000005	0.00220	0.00222	0.000004	0.001806	0.001822
N	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000

Notes: Includes all business applications for the period 2004q3-2003m10. Each column corresponds to estimates from a Linear Probability Model. The dependent variables are: AI All – indicator of an AI business application, AI Main Bus. Line – indicator of AI application based on main business line, AI Bus./Trade Name – indicator of AI business applications based on business/trade name. The coefficient estimates are expressed in percentages. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 4: Expected initial size and application characteristics

	(1)	(2)	(3)	(4)
	Employment	Employment	Employment	Employment
ai (all)	0.2161***		0.0346***	
	(0.0092)		(0.0017)	
ai (main bus. line)		0.2566***		0.0386***
		(0.0133)		(0.0024)
ai (bus./trade name)		0.1715***		0.0303***
		(0.0128)		(0.0023)
wage			3.0235***	3.0235***
			(0.0003)	(0.0003)
corp			0.0195***	0.0195***
			(0.0002)	(0.0002)
new bus./hire			-0.0180***	-0.0180***
			(0.0001)	(0.0001)
prev. ein			-0.0015	-0.0015
			(0.0028)	(0.0028)
exec./trade name			0.0291***	0.0291***
			(0.0002)	(0.0002)
bus. address			0.0150***	0.0150***
			(0.0001)	(0.0001)
year controls	*	*	*	*
state controls	*	*	*	*
week controls	*	*	*	*
industry controls	*	*	*	*
R^2	0.0750	0.0750	0.9632	0.9632
N	28,750,000	28,750,000	28,750,000	28,750,000

Notes: Includes all business applications for the period 2017m4-2003m10 – data for expected max initial employment starts in 2017m4. Each column corresponds to estimates from an OLS regression. The dependent variable is DHS-transformed expected maximum initial employment, winsorized at 100 employees at the top. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), an AI application based on main business line (ai (main bus. line)), and AI applications based on business/trade name (ai (bus./trade name)), high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 5: Transition rate and application characteristics

	(1)	(2)	(3)	(4)
	Employer Business	Employer Business	Employer Business	Employer Business
ai (all)	0.0290***		-0.0040	
	(0.0037)		(0.0035)	
ai (main bus. line)		0.0249***		-0.0128***
		(0.0049)		(0.0047)
ai (bus./trade name)		0.0342***		0.0069
		(0.0058)		(0.0054)
wage			0.2481***	0.2481***
			(0.0002)	(0.0002)
corp			0.0977***	0.0977***
			(0.0002)	(0.0002)
new bus./hire			0.0150***	0.0150***
			(0.0002)	(0.0002)
prev. ein			-0.0126***	-0.0126***
			(0.0005)	(0.0005)
exec./trade name			0.0009***	0.0009***
			(0.0001)	(0.0001)
bus. address			0.0096***	0.0096***
			(0.0002)	(0.0002)
year controls	*	*	*	*
state controls	*	*	*	*
week controls	*	*	*	*
industry controls	*	*	*	*
R^2	0.0888	0.0888	0.1930	0.1930
N	42,580,000	42,580,000	42,580,000	42,580,000

Notes: Includes all business applications for the period 2004q3-2003m10. Each column corresponds to estimates from a Linear Probability Model. The dependent variable is indicator of transitioning to an employer business within 8 quarters of the application. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), an AI application based on main business line (ai (main bus. line)), and AI applications based on business/trade name (ai (bus./trade name)), high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 6: Duration to transition and application characteristics

	(1)	(2)	(3)	(4)
	Duration to Transition	Duration to Transition	Duration to Transition	Duration to Transition
ai (all)	0.0290		0.0679	
	(0.0567)		(0.0548)	
ai (main bus. line)		0.0550		0.1055
		(0.0768)		(0.0751)
ai (bus./trade name)		-0.0002		0.0258
		(0.0838)		(0.0799)
wage			-1.0627***	-1.0627***
			(0.0017)	(0.0017)
corp			0.1883***	0.1883***
			(0.0020)	(0.0020)
new bus./hire			0.2378***	0.2378***
			(0.0024)	(0.0024)
prev. ein			-0.0037	-0.0037
			(0.0046)	(0.0046)
exec./trade name			0.1880***	0.1880***
			(0.0021)	(0.0021)
bus. address			-0.0219***	-0.0219***
			(0.0021)	(0.0021)
year controls	*	*	*	*
state controls	*	*	*	*
week controls	*	*	*	*
industry controls	*	*	*	*
R^2	0.0504	0.0504	0.1220	0.1220
N	5,666,000	5,666,000	5,666,000	5,666,000

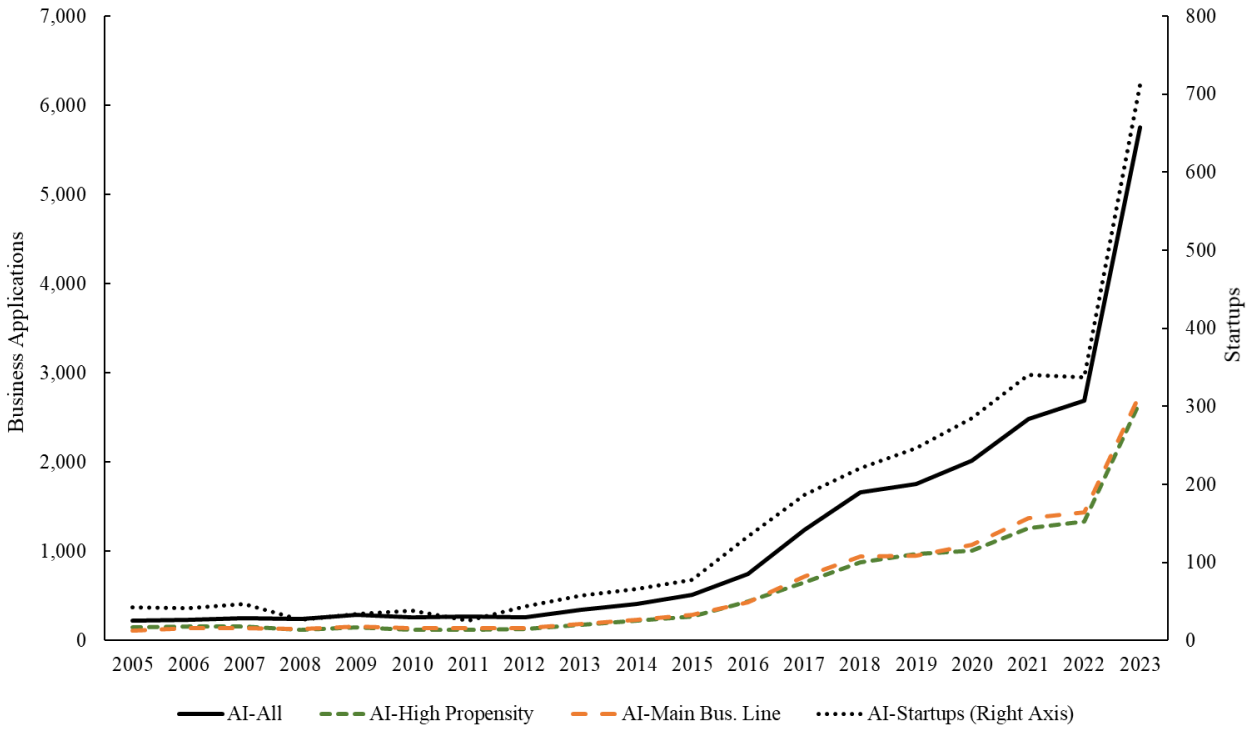
Notes: Includes all business applications for the period 2004q3-2003m10. Each column corresponds to estimates from an OLS regression. The dependent variable is DHS-transformed duration (in quarters) from application to transition. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), an AI application based on main business line (ai (main bus. line)), and AI applications based on business/trade name (ai (bus./trade name)), high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 7: Business outcomes and application characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Employment	Pay	Revenue	Pay/Emp	Revenue/Emp	Pay/Revenue	Exit
ai (all)	0.129***	0.311***	0.097**	0.226***	0.015	0.207***	0.031***
	(0.0276)	(0.0335)	(0.0476)	(0.0196)	(0.0359)	(0.0307)	(0.0058)
wage	0.156***	0.142***	0.076***	-0.002***	-0.049***	0.047***	-0.003***
	(0.0008)	(0.0009)	(0.0010)	(0.0006)	(0.0008)	(0.0007)	(0.0001)
corp	0.046***	0.131***	-0.012***	0.137***	-0.016***	0.108***	-0.022***
	(0.0010)	(0.0011)	(0.0013)	(0.0007)	(0.0010)	(0.0008)	(0.0002)
new bus./hire	0.041***	0.066***	0.061***	0.016***	0.012***	0.020***	0.018***
	(0.0011)	(0.0012)	(0.0014)	(0.0007)	(0.0010)	(0.0008)	(0.0002)
prev. ein	0.065***	0.052***	0.045***	-0.003**	-0.004**	0.011***	0.004***
	(0.0022)	(0.0025)	(0.0025)	(0.0014)	(0.0019)	(0.0015)	(0.0003)
exec./trade name	0.121***	0.141***	0.080***	0.049***	0.021***	0.010***	-0.001***
	(0.0010)	(0.0012)	(0.0013)	(0.0007)	(0.0010)	(0.0008)	(0.0001)
bus. address	0.011***	-0.017***	-0.004***	-0.036***	-0.017***	-0.016***	0.002***
	(0.0010)	(0.0011)	(0.0013)	(0.0007)	(0.0010)	(0.0008)	(0.0002)
BFS year FE	*	*	*	*	*	*	*
LBD year FE	*	*	*	*	*	*	*
industry FE	*	*	*	*	*	*	*
week FE	*	*	*	*	*	*	*
state FE	*	*	*	*	*	*	*
age categories	*	*	*	*	*	*	*
firm size categories				*	*	*	*
R^2	0.180	0.146	0.177	0.270	0.278	0.273	0.030
N	33,500,000	33,500,000	17,200,000	33,500,000	17,200,000	17,200,000	30,700,000

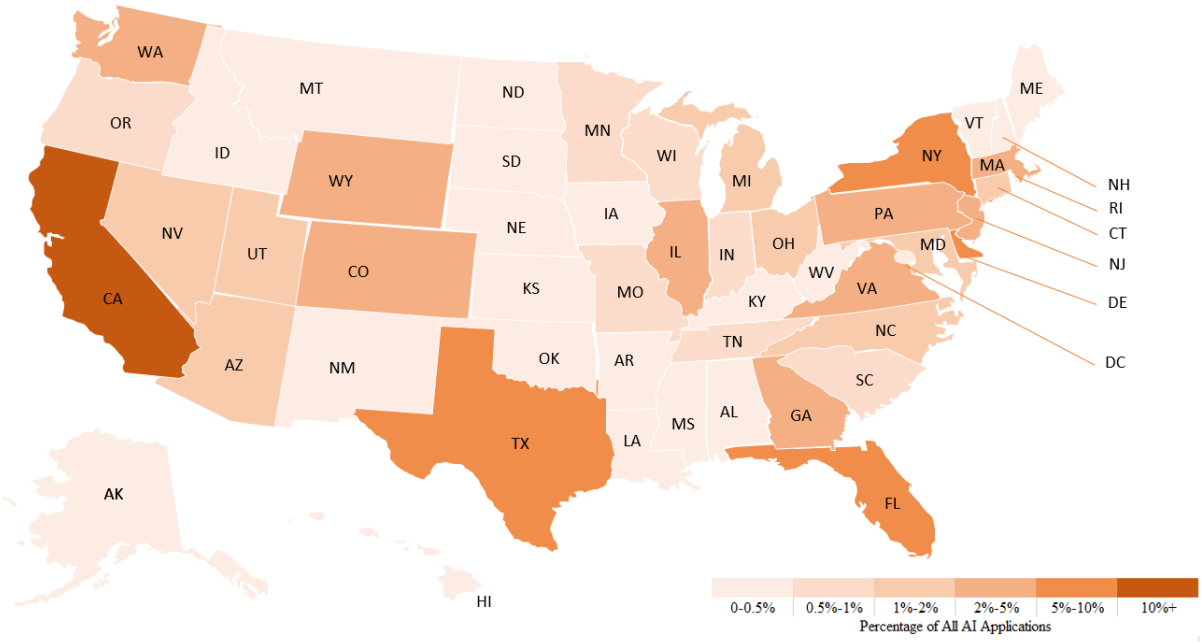
Notes: Includes startups (new employer businesses) from all business applications for the period 2004q3-2003m10. Columns 1-6 provide estimates from an OLS regression conditional on survival – column 7 is based on an LPM. The dependent variables are DHS-transformed except for exit, which is an indicator variable (1 if the business exits in a given year; 0 otherwise). All continuous variables are winsorized in the right tail at 99.9%. The revenue sample is much smaller because of missing revenue for many firms in the LBD – firm-level weights in LBD based on revenue-reporting propensity were used in regressions involving revenue to make the sample representative. Standard errors clustered at the firm level are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). All regressions include fixed effects for application (BFS) year, LBD year, state, application week (of the year), industry (4-digit NAICS). We also control for categories of firm age (measured as LBD year - BFS year). Columns (4)-(7) also control for firm size class. Observation counts (N) are rounded to protect confidentiality.

Figure 1: AI business applications and formations per year



Notes: Uses all AI business applications for the period 2005-2023m10, AI business formations (also referred to as transitions or startups) within 8 quarters of application for the period 2005-2019, and projected AI business formations for the period 2020-2023. Plotted are all AI business applications (AI-All), AI business applications based on main business line (AI-Main Bus. Line), AI business applications based on business/trade name (AI-Bus./Trade Name), and AI employer startups originating from applications within 8 quarters of application (AI-Startups) – uses predicted startups after 2019.

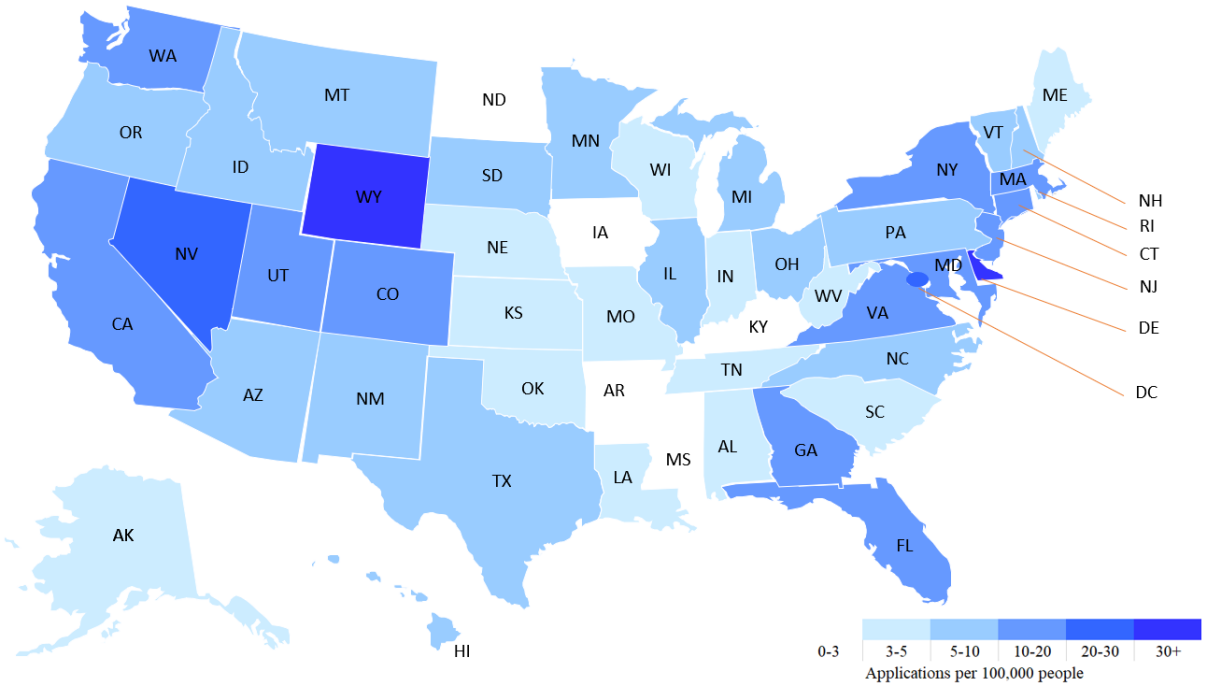
Figure 2: The share (%) of AI business applications by state



Notes: Includes all AI business applications for the period 2004q3-2023m10.

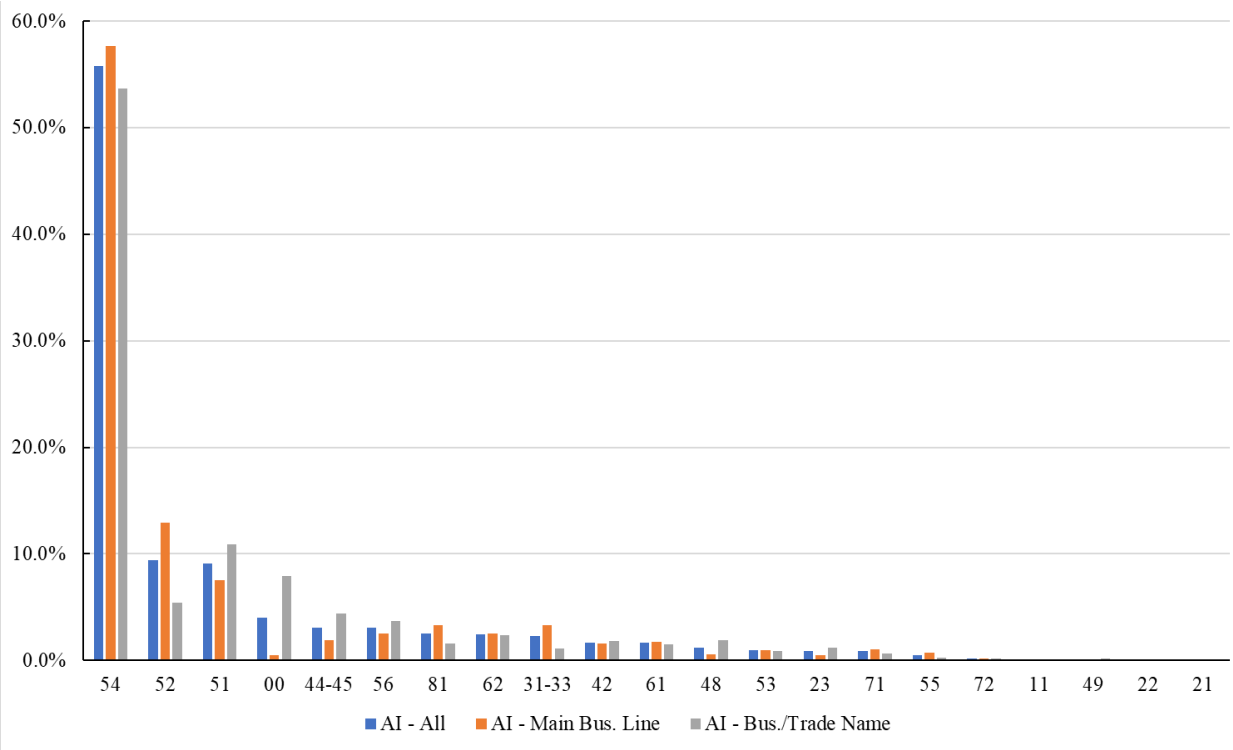
Figure 3: AI business applications per capita by state

(per 100,000 residents of age 18-64)



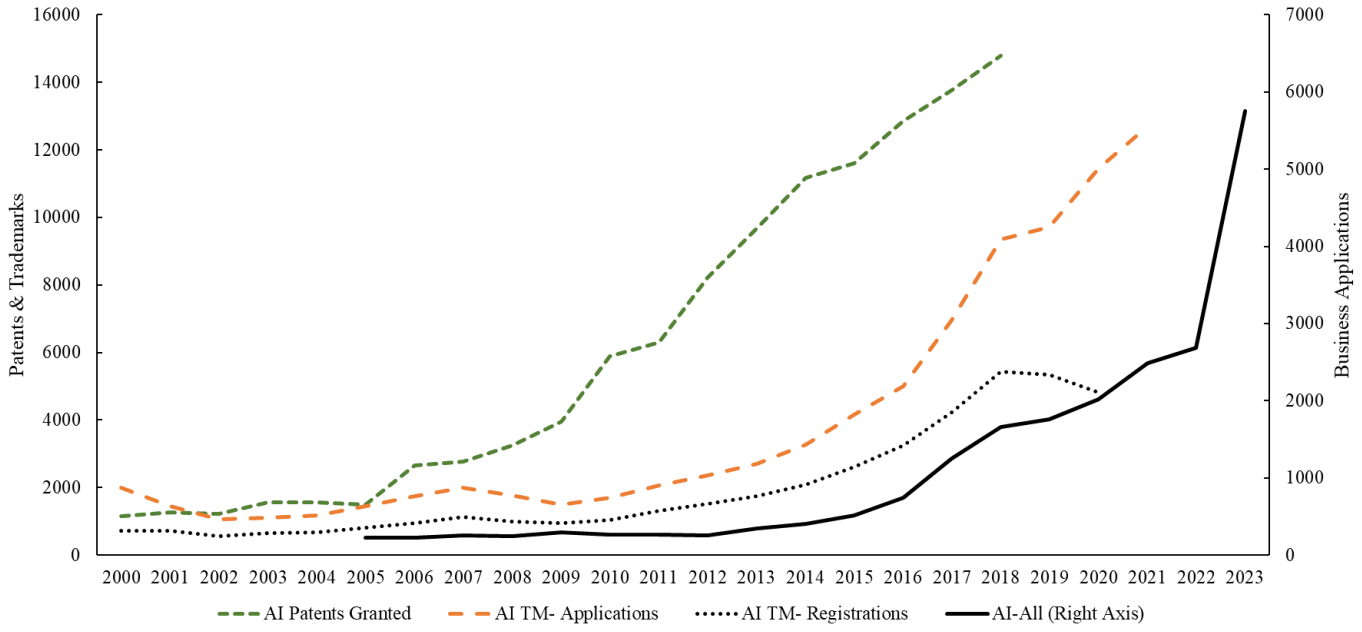
Notes: Includes all AI business applications for the period 2004q3-2023m10. Uses state-level average estimated resident population for ages 18-64 over the period 2005-2022 – estimates are from U.S. Census Bureau.

Figure 4: The share (%) of AI business applications by sector



Notes: Includes all AI business applications 2004q3-2023m10. Sectors are in decreasing order of AI business applications (AI-All) share from left to right. Sector code "00" refers to applications that are not classified into any NAICS sector by the industry assignment algorithm used for BFS. AI-Main. Bus Line and AI-Bus./Trade Name shares for NAICS 21 and 22 are suppressed to protect confidentiality.

Figure 5: AI patents, trademarks, and business applications per year

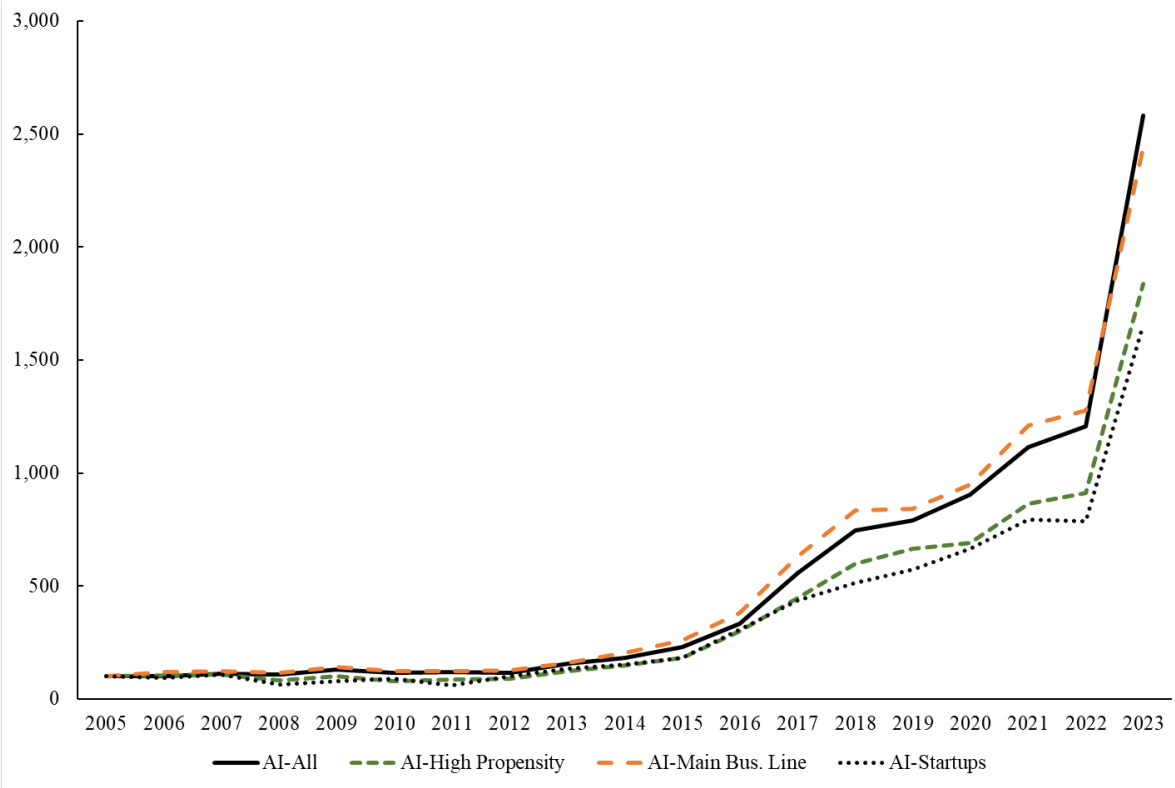


Notes: Includes all AI business applications 2005-2023m10. AI patents are those that are identified by Carnegie Mellon University with 95% confidence. Trademark applications and registrations were identified using text search on United State Patent and Trademark Office's (USPTO) Trademark Case File Database (TCFD).

Appendix

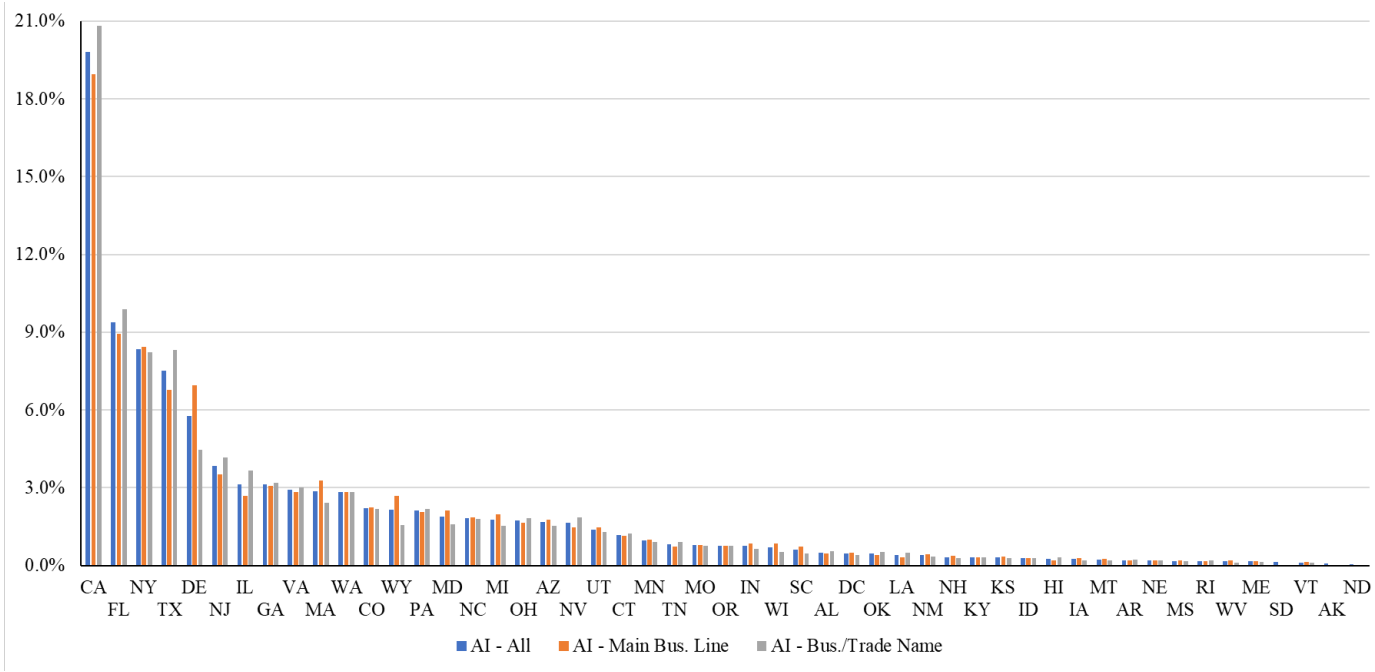
A Additional figures

Figure A.1: AI business applications and formations per year (Index_2005=100)



Notes: Uses all AI business applications for the period 2005-2023m10, AI business formations (also referred to as transitions or startups) within 8 quarters of application for the period 2005-2019, and projected AI business formations for the period 2020-2023. Plotted are all AI business applications (AI-All), AI business applications based on main business line (AI-Main Bus. Line), AI business applications based on business/trade name (AI-Bus./Trade Name), and AI employer startups origination from the applications within 8 quarters of application (AI-Startups) – uses predicted startups after 2019. Each series is indexed to 100 for 2005.

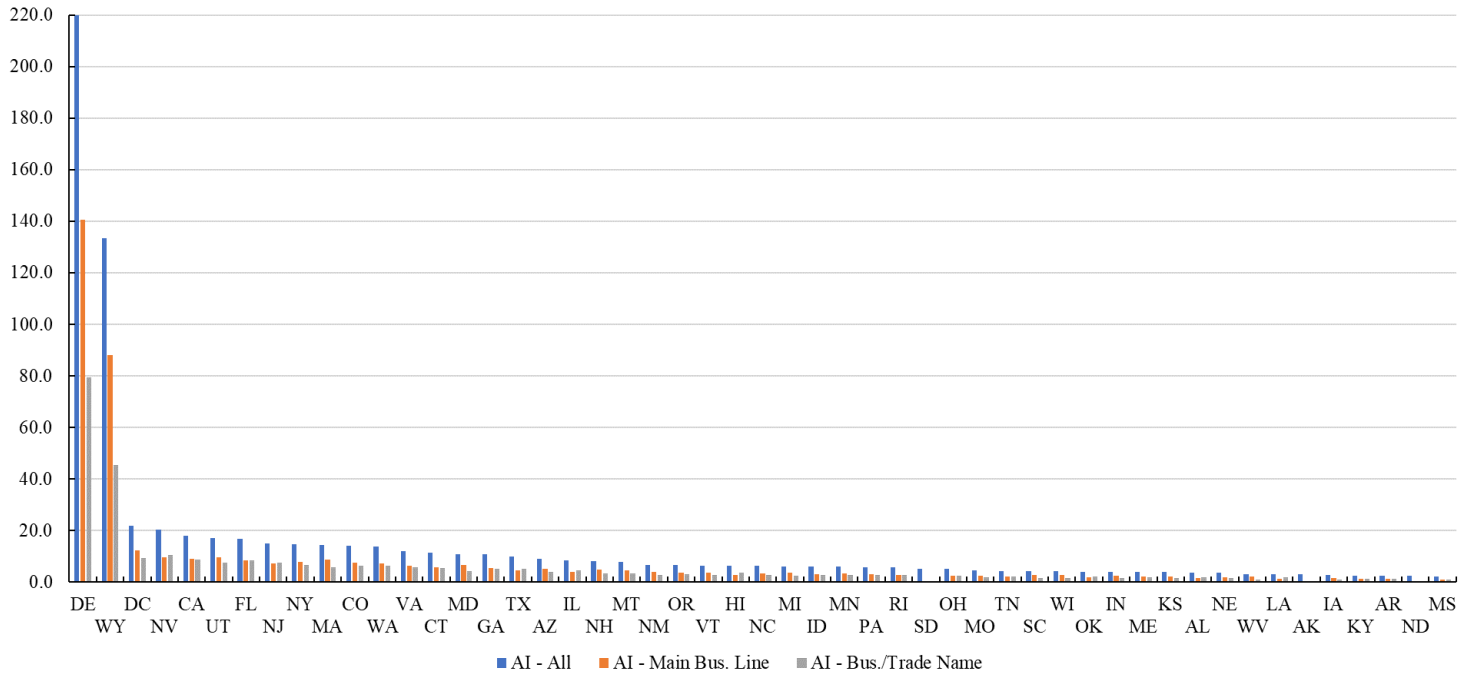
Figure A.2: The share (%) of AI business applications by state



Notes: Includes all AI business applications for the period 2004q3-2023m10. States are in decreasing order of AI business applications (AI-All) share from left to right. AI-Main. Bus Line and AI-Bus./Trade Name shares for AK, ND, and SD are suppressed to protect confidentiality.

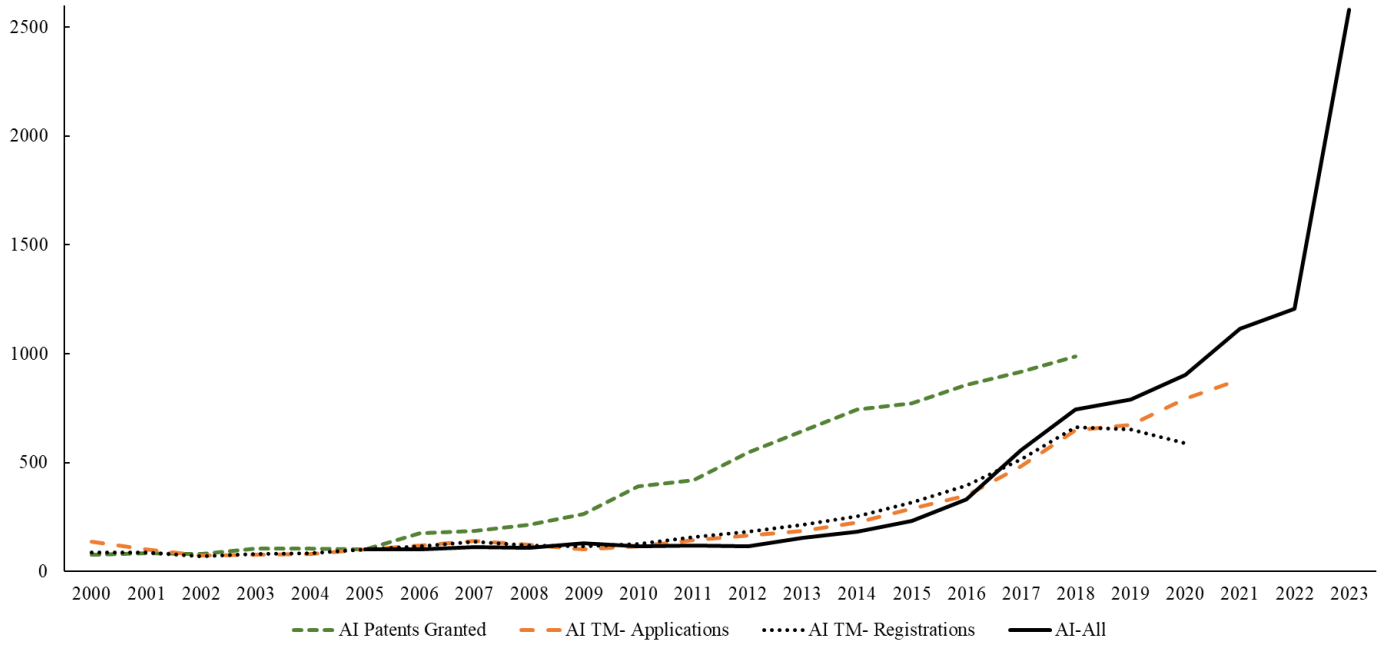
Figure A.3: AI business applications per capita by state

(per 100,000 residents of age 18-64)



Notes: Includes all AI business applications for the period 2004q3-2023m10. Uses state-level average estimated resident population for ages 18-64 over the period 2005-2022 – estimates are from U.S. Census Bureau. States are in decreasing order of AI business applications (AI-All) per capita (per 100,000 residents of age 18-64) from left to right. AI-Main. Bus Line and AI-Bus./Trade Name shares for AK, ND, and SD are suppressed to protect confidentiality.

Figure A.4: AI patents, trademarks, and business applications per year (Index_2005=100)



Notes: All series are indexed to 100 for 2005 to facilitate the comparison of trends with that of AI business applications (AI-All). AI patents are based on the identification by Branstetter et al. (2023). AI trademark applications and registrations are based on text search using USPTO's Trademark Case File Database (TCFD) and AI-related keywords.